

Lucía Cabrera

luciacabrera.ar



Pick a flower on Earth and
you move the farthest star.
– Paul Dirac

Areas of specialization

• Theoretical & Mathematical
Physics • String Theory &
M-Theory • AI Evaluation &
Benchmarking

Languages

Spanish	C2	mother tongue
English	C2	• • • • •
German	B2	• • • • •
Italian	B2	• • • • •

Programming Languages

haskell	=====
python	=====
lean	=====

Dev Stack

Git	=====
Jupyter Notebooks	=====
OpenAI API	=====

Other Interests

• Functional Programming
• Quantitative Research
• Structured AI Reasoning
• Renaissance Music
• Philosophy



EDUCATION

2021 – exp. 2026

PhD Candidate in Physics

BALSEIRO INSTITUTE · Bariloche, Argentina

Thesis: Particle Physics, Quantum Gravity and Dualities in String Theory.

Supervisor: Prof. Gerardo Aldazabal **Co-supervisor:** Prof. Bobby Acharya



2019

Physics Licentiate Degree (equivalent BSc + MSc)

NATIONAL UNIVERSITY OF ROSARIO · Rosario, Argentina

Thesis: Dual Holographic Description of Yang-Mills Plasmas **Supervisor:** Prof. Martin Schwelling (La Plata Institute of Physics - National University of La Plata). **GPA:** 9.53/10



EXPERIENCE

2025 – present

AI Evaluator & Benchmarking Researcher

POINDEXTER LABS · London, United Kingdom

• Designed adversarial math/physics datasets to probe LLM reasoning
• Contributed to AI safety prompting • Analyzed model failure modes &
co-authored an ACL 2026 workshop paper on prompt-length effects.



2022 – 2025

Teaching Assistant

BALSEIRO INSTITUTE · Bariloche, Argentina

• Instructed undergraduate courses in linear algebra, differential equations, and complex analysis.



2016 – 2019

Teaching Assistant

NATIONAL UNIVERSITY OF ROSARIO · Rosario, Argentina

• Instructed undergraduate courses in linear/spectral algebra, complex analysis and distribution theory.



PUBLICATIONS

2026

Eigenforms and special holonomy

arXiv:2608.09878 [math.DG]

B. ACHARYA, L. CABRERA, S. CORBO, J. D. LOTAY

Existence and non-existence results for L^2 eigenforms on complete non-compact Ricci-flat manifolds with special holonomy.

2026

Too long; didn't solve • Accepted to EvalEval Workshop @ ACL 2026

arXiv:2604.07593 [cs.AI]

L. CABRERA, I. SAXTON-KNIGHT, J. D'ARCY

On how prompt length shapes LLM behaviour and discriminativeness.

Upcoming

On capturing stringy features within a field theory description

G. ALDAZABAL, E. ANDRÉS, L. CABRERA, M. MAYO

Embedding stringy features into a QFT and its hidden symmetries.

HIGHLIGHTED SCHOLARSHIPS & AWARDS

2025

ZONTA Women in STEM Award – South American Winner

ZONTA INTERNATIONAL



2023

STEP Fellowship

INTERNATIONAL CENTRE FOR THEORETICAL PHYSICS · Trieste, Italy

International research fellowship supporting three visits to ICTP.



2020

Highest GPA in Physics

PUNTO BIZ MAGAZINE · Santa Fe, Argentina

Highest GPA in a physics degree in the province of Santa Fe, Argentina.



2019

Doctoral Fellowship

CONICET · Argentina

CONICET



2019

Summer Research Scholarship

BALSEIRO INSTITUTE · Bariloche, Argentina

Research internship. **Project:** Communication Servers for LAHN/ANDES.



SELECTED TALKS

2024

Invited Talk • Imperial College London. United Kingdom.

2024

Contributed Talk • 5th Young Frontiers Meeting. Online.

2024

Contributed Talk • 1st Online Conference for Women in Theoretical Physics from LATAM. Online.

RESEARCH VISITS

2025

ICTP - as STEP fellow. Italy.

2024

ICTP - as STEP fellow. Italy.

2023

ICTP - as STEP fellow. Italy.

2022

IPhT Saclay, France.